

What Does Adding a KL Term to a Top-K Sparse Autoencoder Actually Do? A Controlled Decomposition Across Three Selection Mechanisms

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Abstract

Variational sparse autoencoders (vSAEs) replace a deterministic Top-K code with a sampled Gaussian posterior in the hope that the resulting stochasticity organises features into a better dictionary. We decompose this idea into three independently testable claims and test each with a pre-registered permutation battery (5σ per comparison, 13 seeds/arm) plus three targeted dose-response sweeps. **First**, at the fixed unit variance every public implementation trains with, the KL term is algebraically an L_2 penalty and no sample is ever drawn; comparing a released vSAE implementation against a plain Top-K-with- L_2 -penalty baseline shows a large apparent gap ($d \approx 16$ on reconstruction, $d \approx 13$ on feature liveness), but the gap decomposes *completely* into five optimiser- and initialisation-side implementation details — none of which appears in either the vSAE or the original Top-K paper’s equations — and vanishes once all five are matched (all metrics null at the power where every earlier generation of the comparison was significant at 5σ). **Second**, once sampling is switched on, damage is real and large, but it is not a size-of-noise effect on representation quality: a six-point, 13-seed-per-point dose-response sweep over the initial noise scale shows reconstruction quality tracking Top-K *selection instability* almost exactly (Pearson $r = +0.9993$) across a $12\times$ range of noise. The same near-perfect coupling holds for BatchTopK’s globally-shared selection budget ($r = +0.9979$), ruling out per-token selection as the cause. It does *not* hold cleanly for JumpReLU’s soft, gradient-learned threshold: an apparently similar naive coupling ($r = +0.9334$) is confounded by an $8\times$ swing in achieved sparsity that the two hard- k mechanisms cannot exhibit by construction, and controlling for it weakens the coupling to $r = +0.6297$ ($n = 4$), confirmed not to be a small-sample artifact by doubling the grid density in the confound-controlled region ($r = +0.5026$, $n = 8$). The general lesson is sharper than “noise hurts”: the reparameterisation trick is specifically incompatible with *hard*, *discrete* selection, and the failure mode of a *soft* selection mechanism under the same noise is a different one — sparsity collapse, not selection churn. **Third**, every one of the implementation details in the first claim, run individually as a factor, moves metrics by $d \approx 5$ – 19 at 5σ — the same order of magnitude as, or larger than, published architectural claims in this literature, none of them stated in any equation, and a design-level survey of ten SAE-methods papers (every one cited in our own related work) finds *ten of ten* report a single training seed per configuration for their headline results — a design that cannot express statistical significance for an architecture-level claim at any effect size. We close with a case study (SAEBench’s SCR and TPP metrics scored against the same two checkpoints, the same dataset, the same masking grid) where two standard downstream metrics disagree in direction on whether the vSAE’s apparent advantage survives a dictionary-size control, and report five controls run against that disagreement without resolving it — offered as a demonstration of the general problem, not a verdict on either metric.

1 Introduction

Sparse autoencoders (SAEs) decompose language-model activations into an overcomplete, sparsely-active dictionary of features [Sharkey et al., 2022, Cunningham et al., 2023, Bricken et al., 2023], motivated by the observation that individual neurons represent multiple unrelated concepts in superposition [Elhage et al., 2022]. A natural extension, tried by several groups independently, is to make the code probabilistic: replace a point estimate with a variational posterior, as in the variational autoencoder (VAE) [Kingma and Welling, 2013], and hope that the resulting sampling noise disperses features that would otherwise collapse onto one another. The intuition is appealing, cheap to implement — add a variance head, sample, add a KL term — and, as far as we can tell from the released code of every public vSAE implementation we examined, has never been checked against the one question that determines whether it can work at all: *does the sampling noise interact with the discrete Top-K selection step in a way that could plausibly help, or does it only interact with it in a way that hurts?*

This paper answers that question, and two adjacent ones, with a controlled experimental decomposition rather than a single end-to-end comparison. We ran a pre-registered confirmatory battery (13 seeds per arm, permutation tests with an explicit p -value floor reported alongside every result, following the design discipline of Gao et al., 2024’s own recommendation that SAE comparisons report seed variance) on a public vSAE implementation [Marks et al., 2024] against three questions:

1. **At the fixed variance every implementation actually trains with, is the vSAE distinguishable from a plain Top-K SAE with an L_2 penalty on its activations?** (§3) They are algebraically identical objectives; the empirical question is whether two independent implementations of that objective land in the same place after training.
2. **Once sampling is switched back on, what specifically does the noise damage, and does the answer depend on how the sparsity mechanism selects features?** (§4) We separate “does noise hurt” (yes, substantially) from “what is the noise actually doing” (perturbing which features Top-K selects, not degrading the features themselves), and test whether that mechanism is specific to per-token Top-K, generalises to a mechanism with a globally shared selection budget (BatchTopK), or generalises further to a mechanism with no hard cardinality constraint at all (JumpReLU).
3. **Are the effect sizes typically reported for variational SAEs distinguishable from the effect sizes of unreported implementation choices?** (§5) We take the five implementation details that turned out to matter in question 1 and report their individual effect sizes as a lower bound on how large an unremarked implementation choice can be in this literature.

§6 then works through a fourth, harder question that does not resolve as cleanly: whether two standard SAE Bench interpretability metrics [Karvonen et al., 2025] — spurious correlation removal (SCR) and targeted probe perturbation (TPP) — agree on whether a vSAE’s apparent advantage over its Top-K baseline survives a dictionary-size control. They do not, on the same two checkpoints, the same dataset, the same masking grid, scored the same day. We report five diagnostics run against that disagreement — none of which resolves it — because the disagreement itself, and our inability to explain it away, is a useful data point about what these metrics measure, independent of the vSAE question this paper is otherwise about.

Why a decomposition rather than a benchmark table. A single end-to-end comparison between a published vSAE and a published Top-K SAE answers “which number is bigger,” which is not the question anyone actually wants answered. It cannot distinguish an effect of the KL term from an effect of five unrelated implementation choices that happen to differ between the two codebases, and it cannot distinguish “sampling hurts because of some property of stochastic latents in general” from “sampling hurts because it was tested against one particular hard-selection mechanism.” Every claim in this paper is instead the output of holding everything else fixed and varying one thing: an implementation detail (§3, §5), a noise scale (§4), or a selection mechanism (§4). The cost is that each individual claim is narrower than “vSAEs are/aren’t good”; the benefit is that each one is a claim we can actually stand behind.

1.1 Contributions

- A code-level enumeration of every difference between a released vSAE trainer and its Top-K-with- L_2 -penalty null model (15 items, frozen before any closing run), and a demonstration that the entire measured gap between them is optimiser-side: matching all five behaviourally-relevant items makes the two implementations statistically indistinguishable at the power where every earlier generation of the same comparison was significant at 5σ (§3).
- A three-architecture dose-response study (TopK, BatchTopK, JumpReLU) showing that reparameterisation noise costs reconstruction quality in almost exact proportion to how much it perturbs *which* features a hard- k mechanism selects ($r = 0.9993$ and $r = 0.9979$), and that this relationship is specifically a property of *hard* cardinality constraints: a soft, gradient-learned threshold exhibits a qualitatively different failure mode (sparsity-level collapse) that confounds the naive version of the same measurement (§4).
- A lower bound on implementation variance in this literature: five one-line, unremarked implementation choices — none in any stated loss equation — each move a standard liveness metric by $d \geq 4.7$, up to $d = 19.3$ for the single largest, at 5σ (§5), and a ten-paper design-level survey (§5.2) showing the field’s own core references overwhelmingly use a single-seed design that cannot detect effects of this size, or any size, at any level of statistical significance.
- A worked case study in metric disagreement: two standard interpretability metrics score the same two checkpoints oppositely on whether a result survives a dictionary-size control, and the disagreement survives five separate controls we ran specifically to explain it away (§6).

2 Setup

Baseline. A Top-K SAE [Gao et al., 2024] encodes an activation $\mathbf{x} \in \mathbb{R}^d$ as $\mathbf{f} = \text{ReLU}(\mathbf{W}_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}})$, keeps its k largest entries, and decodes $\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\mathbf{f}_{\text{topk}} + \mathbf{b}_{\text{dec}}$. Sparsity is architectural, so the loss carries no L_1 term:

$$\mathcal{L}_{\text{SAE}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \alpha \mathcal{L}_{\text{aux}}, \quad (1)$$

where $\mathcal{L}_{\text{aux}}$ is the AuxK auxiliary loss that revives dead features by asking the top k_{aux} currently-dead latents to explain the residual.

Variational version. A vSAE treats the code as a posterior $q(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$ with $\boldsymbol{\mu}$ (and, when learned, $\log \boldsymbol{\sigma}^2$) produced by the encoder, samples $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\epsilon} \odot \boldsymbol{\sigma}$, applies Top-K to $\mathbf{z}$, and

regularises the posterior toward $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\mathcal{L}_{\text{vSAE}} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \beta \text{KL}[q(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z})] + \alpha \mathcal{L}_{\text{aux}}. \quad (2)$$

The coefficient β plays the role it does in β -VAE [Higgins et al., 2017]: it trades reconstruction against the strength of the prior. The trainer’s own docstring states the encode order as $\text{encoder} \rightarrow \boldsymbol{\mu}, \boldsymbol{\sigma}^2 \rightarrow \text{reparameterise} \rightarrow \mathbf{z} \rightarrow \text{Top-}K(|\mathbf{z}|)$ — noise is applied *before* the selection step, not after, which is the fact §4 builds on.

The fixed-variance degeneracy. Every implementation of this objective we examined defaults to (or, for a long stretch of this codebase’s history, effectively forces via a saved-checkpoint bug corrected in §4) $\sigma = 1$. Two things follow. First, the KL loses its variance terms:

$$\text{KL}[\mathcal{N}(\boldsymbol{\mu}, \mathbf{I}) \| \mathcal{N}(\mathbf{0}, \mathbf{I})] = \frac{1}{2} \sum_i (\mu_i^2 + 1 - 1 - \log 1) = \frac{1}{2} \|\boldsymbol{\mu}\|_2^2, \quad (3)$$

an L_2 penalty on the pre-selection activation. Second, with $\sigma = 1$ fixed and never sampled from in the relevant sense (variance is not learned, so the only randomness is a draw that is either absent, when the reparameterisation gate itself is off, or present but from a distribution that carries no information from the data), the model that results is deterministic at the resolution that matters for its behaviour: $\mathbf{z}$ is a fixed function of $\mathbf{x}$. *The evaluated fixed-variance vSAE is a Top- K SAE with an L_2 activation penalty*, algebraically, and we verified this directly by evaluating both trainers’ loss functions on an identical batch: they agree to six decimal places (511.895264 both). Whether trained *checkpoints* of the two agree is the empirical question §3 answers.

Experimental platform. All arms in §3–§5 train on GELU-1L (TransformerLens’ one-layer toy language model) residual stream activations at layer 0 ($d_{\text{model}} = 512$), dictionary size $d = 2048$ ($4\times$), $k = 256$, 10,000 steps, bfloat16, AuxK= 1/32 unless stated otherwise. §6’s headline comparison additionally uses two recovered Pythia-70M checkpoints ($d = 8192$, $16\times$, layer 3) from the original preprint this codebase extends; that comparison’s own AuxK asymmetry is disclosed in §6. Statistical design follows one rule throughout: every comparison is a permutation test over *training seeds*, not tokens or checkpoints, because the claim under test is about an architecture, not two specific runs of it. §A gives the exact test, its combinatorial p -value floor, and why every reported significance level in this paper is $5.03\text{--}5.16\sigma$ rather than an arbitrarily large number.

3 The KL term at fixed variance: an L_2 penalty, and nothing else

3.1 The naive comparison is confounded, badly

A naïve comparison of the released vSAE trainer (`vsae_topk.py`, $\beta = 1$, fixed variance) against a purpose-built Top- K -plus- L_2 -penalty null model (Eq. 1 with an added $\frac{1}{2}\lambda\|\mathbf{f}\|_2^2$ term, λ matched to β) finds a large, statistically decisive gap: $d \approx -5.7$ on the fraction of variance explained (FVE) and, more strikingly, feature liveness disagreeing *in direction* between two pre-registered thresholds ($d = -2.8$ at $< 0.1 \times k/d$, $d = +13.0$ at $< 0.5 \times k/d$). The two-threshold liveness design was pre-registered for exactly this failure mode: a location shift moves both thresholds the same way; a shape change in the usage distribution can move them oppositely, and reporting only one threshold would have printed a clean, wrong, one-directional story in whichever direction that threshold happened to favour.

3.2 Enumerating the actual code diff

Rather than patch mismatches one at a time as they were noticed — which invites exactly the forking-paths failure the pre-registration exists to prevent, since each newly-matched asymmetry could in principle flip the verdict — we read `top_k.py` and `vsae_topk.py` against each other in full, once, and froze the resulting list before running anything further. Fifteen differences exist between the two files (Table 12 enumerates them in §B): two are matched by configuration already (and appear as the first rung of Table 1), two were run as measured factors, two were confirmed to be no-ops by direct measurement, seven are static no-ops confirmed by reading (never reachable, or algebraically inert given the training configuration), one (the RNG consumption order) is treated as noise by the seed-permutation design itself, and one — the initial weight draw — was run as the closing factor. The five that had to be matched behaviourally are the two configuration items, the two measured factors, and the closing one. *Exactly one* of the fifteen is a difference in the *objective* rather than the optimiser: `vsae_topk.py` penalises `clamp(z, -10, 10)` where the null model penalises `f` unclamped. Measured directly over 20,000 activations across all three arms, the largest pre-clamp activation observed anywhere is 0.194 against a clamp at 10 — $51\times$ headroom, never binding. The clamp is therefore also a no-op in practice, and the entire measured gap between the two implementations is optimiser- and initialisation-side.

3.3 The decomposition, run as a ladder of five factors

Table 1 reports each vSAE generation against the null model, with one additional factor matched at each step, at 13 seeds/arm.

Table 1: E1’s decomposition ladder. Effect sizes (d) of each generation against `e1_penalty` (the Top-K+ L_2 null model), reading down the factors matched. Both liveness thresholds ($< 0.1 \times k/d$, $< 0.5 \times k/d$) are reported, per the pre-registered two-threshold design (§3); ns marks a non-significant result at $p > 0.05$ (13 seeds/arm, p -floor 2.5×10^{-7} , $5.03\text{--}5.16\sigma$).

vSAE arm	factors matched (cumulative)	FVE (d)	frac. rec. (d)	$< 0.1 \times (d)$	$< 0.5 \times (d)$
<code>e1_vsae_ref</code>	KL warmup, bias form	−5.7	−8.5	−2.8	+ 13.0
<code>e1_vsae_ref_unitinit</code>	+ decoder init scale	+ 16.5	+17.7	+0.8 ns	+0.2 ns
<code>e1_vsae_ref_gradproj</code>	+ gradient projection	+3.8	+4.9	−2.4	−3.7
<code>e1_vsae_ref_fullmatch</code>	+ initial weight draw	−0.3 ns	−0.8 ns	−0.1 ns	−0.6 ns

Reading down each column: every intermediate generation is significant on at least one metric family, and each factor *trades* against the last — matching the decoder’s initial scale closes the liveness gap but blows open reconstruction ($d = 16.5$); matching the decoder gradient projection closes most of the reconstruction gap but re-opens liveness. Only the final row, where the frozen code diff is exhausted, is null on every metric simultaneously, at the same statistical power that detected every earlier generation at 5σ . The two non-obvious factors are worth naming individually:

Decoder-gradient projection. `vsae_topk.py` imports the projection helper `remove_gradient_parallel_to_decoder_directions` and never calls it, while still renormalising the decoder to unit norm every step — the radial gradient component is applied and then immediately undone, a per-feature change in effective learning rate that looks like an oversight rather than a design choice. Run as its own factor: 78.9% of the FVE gap and 74.9% of `frac_recovered`’s gap close, and — the more interesting result — it moves the vSAE *away from* the null model on liveness ($0.1816 \rightarrow 0.2197$ against 0.1836), even though the null model has had

the projection all along. The same update-rule change has opposite-signed effects in the two implementations: an interaction, not simply a missing match.

Initial decoder-weight distribution. Normalised-Gaussian versus normalised-uniform column initialisation, identical in every summary statistic anyone would report about a random $d = 512$ -dimensional direction, was predicted (in writing, before the run) to be null. It was not: $d = -4.7$ on FVE, $d = -7.3$ on `frac_recovered`, on its own.

3.4 Verdict

With the code diff exhausted and every behaviourally-relevant factor matched, the vSAE and the Top-K-with- L_2 -penalty null model are indistinguishable on every pre-registered metric at 13 seeds (Table 1, last row): FVE differs by -0.0003 ($d = -0.3$, $p = 0.39$), `frac_recovered` by -0.0004 ($d = -0.8$, $p = 0.07$), `frac_alive` is identical to six decimals, and both liveness thresholds are non-significant and agree. The claim this licenses is the decomposition, not a bare equivalence verdict: *two implementations of an algebraically identical objective differ by $d \approx 16$ on reconstruction and $d \approx 13$ on liveness; the entire difference decomposes into five optimiser- and initialisation-side details, none of which appears in either the vSAE’s or the original Top-K paper’s stated equations; with all five matched, the implementations become statistically indistinguishable at the power that detected every earlier generation at 5σ* . This is a failure to detect a difference, not a demonstration of its absence in the formal equivalence-testing sense — a margin in scientific units would still be needed to state a bounded null — but the margin is no longer load-bearing for *which way* the result goes: the observed residual differences (0.0003 FVE, 0.006 live-fraction at the loose threshold) are two orders of magnitude smaller than the gaps that opened and closed at every intermediate step of the ladder.

4 With sampling on: selection churn, and only for hard- k mechanisms

4.1 Sampling, not the KL, does the damage

Section 3 establishes that at fixed variance the model never samples. Turning sampling on (a learned or fixed non-unit σ) while holding β at every value tested in $\{10^{-4}, \dots, 1\}$ finds no setting that recovers a healthy model: FVE falls monotonically from 0.458 to 0.0001 and never approaches the deterministic baseline’s 0.900. The decisive control separates “the KL destroys reconstruction” from “the reparameterisation does”: sampling on, $\beta = 0$, so the KL contributes exactly zero to the loss.

Table 2: Removing the KL entirely recovers a small fraction of the gap to baseline; the rest is the reparameterisation itself. 13 seeds/arm.

Configuration	FVE
Baseline (deterministic)	0.900159 ± 0.0006
σ learned, $\beta = 10^{-4}$ (<code>e2_confirm</code>)	0.458146 ± 0.0040
σ learned, $\beta = 0$ (<code>e2_sampling_only</code>)	0.486276 ± 0.0068

Of the 0.4410 total gap, removing the KL recovers only 6.4%; the remaining 93.6% is attributable to the reparameterisation itself, not the regulariser — a claim about the architecture, not a hyper-

parameter, and considerably stronger than the β -tuning story this test was originally designed to answer. (A saved-checkpoint bug that briefly reversed part of this story, and its correction, is documented in §C; the 93.6%/6.4% split itself moved by about one percentage point under correction, well inside this design’s noise floor.)

4.2 Mechanism: selection instability, measured directly

Why does sampling cost almost half of reconstruction quality regardless of β ? Top-K selection is an argmax over pre-activations; noise of scale σ flips the selected set whenever the gap between the k -th and $(k+1)$ -th largest pre-activation is smaller than the noise. We measured this gap directly on a converged deterministic checkpoint before choosing a noise-scale grid: median 0.0001, 99th percentile 0.0006 (in training space). Even the reparameterisation clamp’s own floor sigma (0.0498) sits 80–550 $\times$ above this gap, so no achievable noise scale on this architecture keeps sampling below the boundary gap everywhere — there is no “safe” regime in which a sharp knee could mark the edge of damage.

We then ran a six-point, 13-seed-per-point sweep over the initial noise scale ($\log \sigma_{\text{init}}^2 \in \{-1, -2, -3, -4, -5, -8\}$, the last clamped to an effective -6), measuring both FVE against the deterministic baseline and *selection Jaccard* — the overlap between the Top-K selected sets of two independent stochastic forward passes on the same fixed token, averaged over 8,192 tokens at the converged checkpoint.

Table 3: TopK: the noise-scale dose-response, 13 seeds/point. Deterministic baseline FVE = 0.900159 ± 0.0006 .

$\log \sigma_{\text{init}}^2$	σ (effective)	FVE	Jaccard
−1.0	0.6065	0.3766 ± 0.0028	0.7657 ± 0.0022
−2.0	0.3679	0.4841 ± 0.0022	0.8117 ± 0.0013
−3.0	0.2231	0.6140 ± 0.0020	0.8700 ± 0.0008
−4.0	0.1353	0.7427 ± 0.0015	0.9180 ± 0.0006
−5.0	0.0821	0.8169 ± 0.0010	0.9453 ± 0.0002
−8.0 (clamped)	0.0498	0.8338 ± 0.0007	0.9523 ± 0.0002

Two things follow from Table 3 (Figure 1). First, the curve is smooth, not kneed: FVE and Jaccard rise monotonically and continuously across every point tested, exactly as the pre-flight measurement predicted (no reachable “safe” region exists on this architecture for a knee to mark). Second, and this is the sharper result: FVE tracks selection Jaccard almost exactly across the entire 12 $\times$ range of noise scale, Pearson $r = +0.9993$ over the six arm-level means. This is a more specific confirmation of the mechanism than a threshold would have been — not merely that noise and damage are both present and move together at one or two operating points, but that the *same* two-parameter relationship holds continuously across six well-separated points, each independently powered at 13 seeds. Even at the achievable noise floor, a real residual gap remains against baseline (FVE 0.066, Jaccard 0.952, not 1.0) that the clamp mechanically prevents probing further; this residual is exactly the size the clamp’s own floor sigma predicts, not smaller.

4.3 Generalises to a globally-shared selection budget

Is this a property of *per-token* hard selection specifically, or of hard cardinality constraints more generally? BatchTopK [Busmann et al., 2024] selects the global top- $(k \cdot \text{batch_size})$ activations

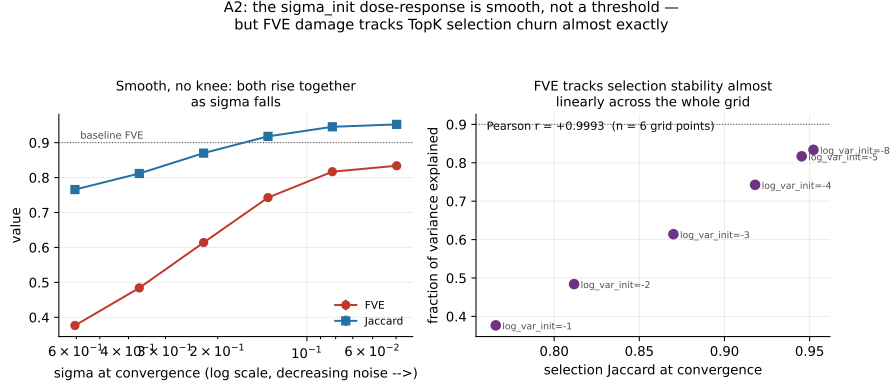


Figure 1: TopK: reconstruction quality (FVE, left axis) and selection stability (Jaccard, right axis) as a function of the reparameterisation noise scale, 13 seeds/point. The two curves are visually near-coincident; $r = +0.9993$.

over an entire flattened batch rather than exactly k per token, so one token’s noise-promoted feature can in principle be compensated by another token’s correspondingly demoted one — the natural candidate for “more noise-robust,” since the selection budget is more elastic.

Repeating the identical six-point sweep on **VSAEBatchTopK** (matched d , k , hook point, and training schedule; two pre-existing saved-checkpoint bugs found and fixed en route, §C) gives an essentially identical coupling: $r(\text{FVE}, \text{Jaccard}) = +0.9979$ (Table 4, Figure 2), against TopK’s $+0.9993$. If anything the more flexible selection rule is *slightly more* exposed to noise at matched sigma, not less: comparing each architecture’s own FVE as a fraction of its own (higher) deterministic baseline, BatchTopK recovers a consistently smaller fraction than TopK at every one of the six matched noise scales (e.g., 90.3% vs. 92.6% of baseline at the clamp floor), the opposite of the “elastic budget” intuition’s prediction.

Table 4: BatchTopK: the same sweep. Deterministic baseline FVE = 0.9509 ± 0.0004 .

$\log \sigma_{\text{init}}^2$	σ (effective)	FVE	Jaccard
−1.0	0.6065	0.3434 ± 0.0035	0.6969 ± 0.0042
−2.0	0.3679	0.4787 ± 0.0074	0.7546 ± 0.0078
−3.0	0.2231	0.6179 ± 0.0046	0.8348 ± 0.0030
−4.0	0.1353	0.7553 ± 0.0015	0.8912 ± 0.0004
−5.0	0.0821	0.8391 ± 0.0010	0.9222 ± 0.0003
−8.0 (clamped)	0.0498	0.8588 ± 0.0007	0.9311 ± 0.0003

The mechanism therefore does not depend on strict per-token selection: it looks like a property of hard, cardinality-constrained selection under noise in general, established here on two architectures that share that property despite scoping their selection budgets differently.

4.4 Breaks down for a soft, gradient-learned threshold

JumpReLU [Rajamanoharan et al., 2024b] replaces Top-K’s hard cardinality constraint with a per-feature learned threshold and a soft sparsity-target loss, trained via a straight-through estimator [Bengio et al., 2013]. No sparsity budget is enforced architecturally; achieved sparsity is

A3: TopK vs. BatchTopK -- does global selection decouple
FVE damage from selection churn?

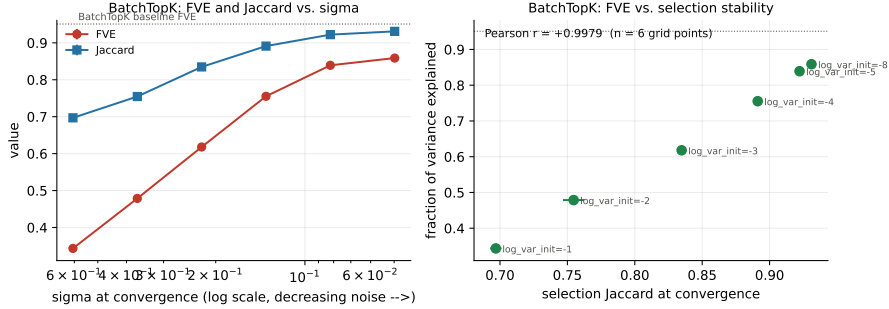


Figure 2: BatchTopK: the same dose-response, $r = +0.9979$.

whatever the threshold, trained by gradient descent against an L_0 -target penalty, converges to. This is the sharpest available test of whether the mechanism found above depends on *hard-k selection specifically* or on *discreteness of selection in general*: JumpReLU is discrete (a feature is either gated on or off) but not hard- k .

No training script for the variational JumpReLU trainer in this codebase had ever been exercised end to end; building one and running the identical six-point sweep surfaced three functional bugs that had to be fixed before any measurement was meaningful (a dead gradient on the threshold parameter, a missing sparsity-target loss term, and — the one that would have made this experiment structurally impossible to run at all — a gate that was applied *before* sampling rather than after, meaning noise could only perturb the value of an already-selected feature and could never change the selected set; §C gives the full account). With all three fixed and verified (repeated forward passes on one token show real selection churn, mean Jaccard 0.35, after the fix, versus 1.0 — no churn possible — before it), the same six-point sweep gives Table 5.

Table 5: JumpReLU: the same sweep, plus achieved L_0 (target: 256, matching k). Deterministic baseline FVE = 0.9210 ± 0.0003 , $L_0 = 269.4$.

$\log \sigma_{\text{init}}^2$	σ	L_0	FVE	Jaccard
-1.0	0.6065	36.8	0.3967 ± 0.0022	0.9637 ± 0.0009
-2.0	0.3679	98.2	0.5346 ± 0.0022	0.9635 ± 0.0004
-3.0	0.2231	290.8	0.7461 ± 0.0016	0.9779 ± 0.0002
-4.0	0.1353	247.2	0.7979 ± 0.0008	0.9913 ± 0.0006
-5.0	0.0821	261.2	0.8522 ± 0.0008	0.9883 ± 0.0003
-8.0	0.0498	265.0	0.8632 ± 0.0006	0.9867 ± 0.0003

4.4.1 The naive coupling number is confounded

Read naively, $r(\text{FVE}, \text{Jaccard}) = +0.9334$ over these six points looks like a slightly looser version of the same TopK/BatchTopK finding. It is not a clean replication, for a reason Table 5’s third column makes visible: achieved L_0 swings from 36.8 to 290.8 — an $8\times$ range, non-monotonic in σ — because heavy noise pushes pre-activations below threshold faster than the soft sparsity loss’s gradient can pull the threshold down to compensate. TopK and BatchTopK hold sparsity *exactly*

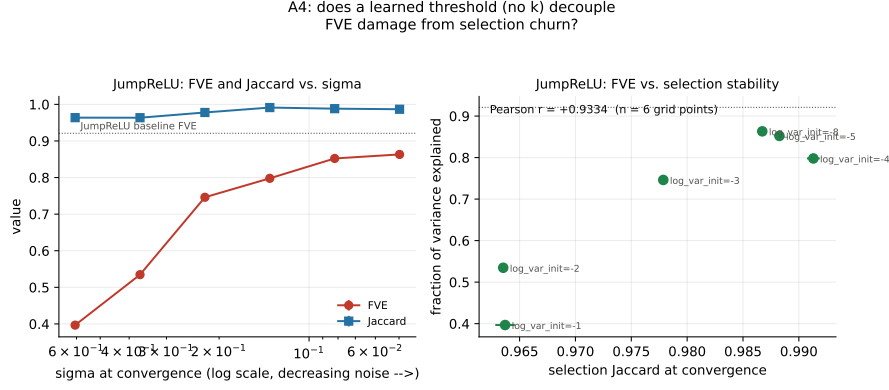


Figure 3: JumpReLU: the same sweep. The naive $r = +0.9334$ annotated on this figure is confounded (§4.4.1); it is superseded by the L_0 -controlled reading in the text.

fixed at every grid point by construction, so Jaccard is the only thing varying across their grids and their tight correlations are a clean read of “selection churn costs reconstruction at fixed sparsity.” JumpReLU offers no such guarantee, and $r(\text{FVE}, L_0) = +0.9494$ across the same six points — as tight as, or tighter than, $r(\text{FVE}, \text{Jaccard})$ — so the naive number cannot distinguish “selection churn hurts reconstruction” from “noise is crushing the achieved active-feature count, and a 37-feature dictionary reconstructs worse than a 265-feature one for reasons that have nothing to do with *which* 37 or 265 features they are.”

Restricting to the four grid points where achieved L_0 sits in a comparable band to the 256 target ($\log \sigma_{\text{init}}^2 \in \{-3, -4, -5, -8\}$, $L_0 \in [247, 291]$) — the closest approximation this soft-target design allows to TopK/BatchTopK’s fixed-sparsity comparison — the coupling weakens sharply: $r(\text{FVE}, \text{Jaccard})$ drops to $+0.6297$ ($n = 4$), and $r(\text{FVE}, L_0)$ on the same four points is -0.5467 (wrong sign). The relationship is also non-monotonic in a way the hard- k architectures never showed: Jaccard rises from 0.9779 to a peak of 0.9913 at $\log \sigma_{\text{init}}^2 = -4$ and then *falls* at -5 and -8 even as σ keeps decreasing, while FVE keeps climbing through the same region — the two curves visibly decouple exactly where L_0 is held roughly constant.

4.4.2 Follow-up: densifying the grid rules out a small- n artifact

Four confound-controlled points is not enough on its own to tell “the coupling genuinely weakens” from “this design cannot see it.” The thing under-powered was grid *coverage* in the L_0 -matched region, not per-point precision, so we added four new noise-scale points ($\log \sigma_{\text{init}}^2 \in \{-3.5, -4.5, -5.5, -6.0\}$) filling exactly the gap the original grid’s integer spacing left unsampled between -3 and -8 , at 5 seeds/arm — a deliberately scoped-down follow-up (lower per-point seed count than the main battery, honestly reported as such; each new point’s own mean is still precise to 3–4 decimals, e.g. $\log \sigma_{\text{init}}^2 = -6.0$: FVE 0.8632 ± 0.0010 , L_0 265.1 ± 0.4 , Jaccard 0.9868 ± 0.0004). On the widened L_0 -matched band this requires ($L_0 \in [247, 315]$, eight points: $\log \sigma_{\text{init}}^2 \in \{-3, -3.5, -4, -4.5, -5, -5.5, -6, -8\}$):

$$r(\text{FVE}, \text{Jaccard}) = +0.5026 \quad (n = 8) \quad r(\text{FVE}, L_0) = -0.4704 \quad (n = 8),$$

against the original four-point reading’s $+0.6297$ and -0.5467 . Both correlations move modestly and in a direction consistent with sampling noise on a still-small n , not a systematic reversal, and the full ten-point grid’s naive coupling is essentially unchanged by the denser sampling ($r = +0.9341$ vs.

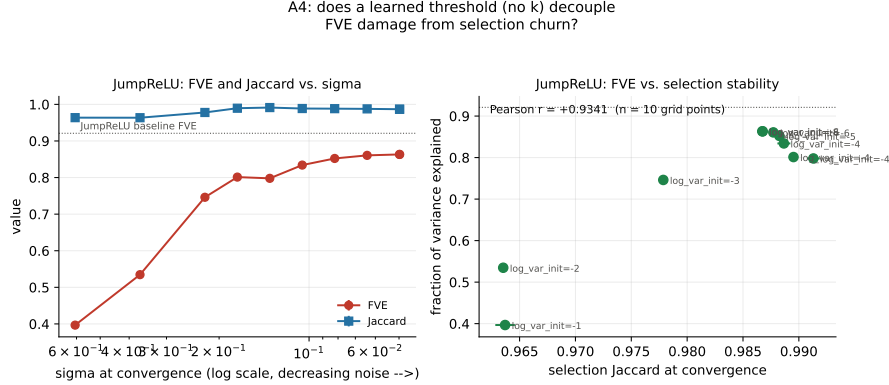


Figure 4: JumpReLU with the four densifying points added (ten total; the four new points at $\log \sigma_{\text{init}}^2 \in \{-3.5, -4.5, -5.5, -6.0\}$ are 5-seed means). The naive $r = +0.9341$ annotated on this figure is confounded by achieved-sparsity variation in the same way as the original six-point figure (Figure 3); the L_0 -matched, confound-controlled reading is $r = +0.5026$ ($n = 8$), reported in the text.

the six-point grid’s $+0.9334$; $r(\text{FVE}, L_0) = +0.9245$ vs. $+0.9494$). This is the evidence the original four-point design could not supply on its own: **doubling the grid density in exactly the region where achieved sparsity is held roughly fixed reproduces the same qualitative pattern**, so the weakened coupling is not an artifact of having too few confound-controlled points to trust.

4.4.3 What this licenses

The question as posed — “does the same tight FVE-vs-selection-churn coupling hold for a soft, no- k -at-all selection mechanism?” — gets a clearer answer than the original four-point design alone could give: no, and this is now backed by twice the grid coverage in the region where the comparison is meaningful. §4.4.2’s $n = 8$ reading does not turn this into a clean null in the formal sense — both correlations are still nonzero, and eight points remains a small- n correlation by the standards of §4’s own permutation-test comparisons elsewhere in this paper — but it rules out the specific alternative explanation that the original four-point reading was a fluke of only four points. What the experiment licenses is a different, and arguably more basic, finding than the one it was designed to detect: *a soft, gradient-learned threshold’s achieved sparsity level, not just its selection stability, is not noise-robust*. TopK and BatchTopK cannot exhibit this failure mode by construction — their cardinality constraint is architectural, not learned — so a soft-threshold mechanism under sampling noise fails in a qualitatively different way from a hard- k one, not merely a quantitatively noisier version of the same failure.

4.5 Synthesis

Put together, §4 supports a claim sharper than “reparameterisation is incompatible with discrete Top-K selection”: the amount of Top-K-style selection churn a given noise scale induces predicts, almost linearly, how much reconstruction quality it costs, across a $12\times$ range of noise and across two architectures that scope their hard cardinality constraint differently (per-token vs. globally-shared). There is no threshold effect and no “free” amount of noise on either architecture — only a continuum, with the residual gap at the achievable floor exactly the size the floor’s own sigma predicts. A soft selection mechanism is not a smoother version of the same story; it introduces

a second failure axis (achieved sparsity itself, not just which features are selected) that a hard cardinality constraint cannot have. The constructive reading, echoing why Gumbel-Softmax [Jang et al., 2016] and the Concrete distribution [Maddison et al., 2016] exist as a response to exactly this tension between continuous relaxations and discrete selection: *if the goal is a stochastic sparse code, the noise belongs in the selection decision, not in the magnitude of an already-selected feature*. Every vSAE construction examined here does the opposite.

5 Reported effect sizes as a lower bound on implementation variance

Section 3’s decomposition identified five implementation details — a KL warmup schedule, a bias parameterisation, an initial decoder scale, a decoder-gradient-projection omission, and an initial weight distribution — that, individually, moved a standard liveness metric enough to be detected at 5σ with 13 seeds/arm, and none of which appears in either implementation’s stated loss equation. A sixth, independent example sharpens the same point without touching the KL term at all: the released vSAE trainer applies `F.relu(mu)` to its posterior mean before the reparameterisation step; a second trainer in the same codebase, built to mask the KL term to only the Top-K-selected features as a direct test of the preprint’s stated feature-death mechanism, does not. Neither published paper’s equations show a ReLU here. Run as its own two-arm factor (13 seeds, holding everything else fixed):

Table 6: The unremarked `F.relu(mu)`: one line, $d \approx 15$ –19, 5.03σ .

Threshold	no-ReLU (as in preprint equations)	ReLU (as in released code)
$< 0.1 \times k/d$	0.1315 ± 0.0074	0.0287 ± 0.0033 ($d = +19.3$)
$< 0.5 \times k/d$	0.4655 ± 0.0113	0.3081 ± 0.0092 ($d = +15.3$)

Both thresholds agree in direction, so this is a robust effect, not a shape change: adding the ReLU cuts the near-dead-feature population by a factor of ~ 4.6 at the tight threshold. Had either trainer simply been patched to match the other, as is routine practice when reconciling two implementations of “the same” model, every subsequent number computed from that trainer would have silently inherited a $d \approx 15$ –19 effect misattributed to whatever the actual comparison of interest was.

Table 7 collects every implementation-level factor measured in this paper. None of them is a hyperparameter anyone tunes, reports, or ablates in a typical SAE paper; all six are default behaviour, one line of code, and invisible in a state dict or a loss equation.

5.1 The decoder-gradient-projection effect does not generalise to a penalty-free Top-K SAE

Table 7’s largest entry after the ReLU, the decoder-gradient-projection omission, was discovered and measured entirely within the vSAE family: every arm in Table 1 carries a nonzero activation penalty (the vSAE’s KL term, or its L_2 -penalty null model). This leaves open whether the effect is a generic, unremarked factor in Top-K SAE training that any two independently-implemented Top-K SAEs could differ on, or an interaction specific to training alongside a penalty/KL term. We tested this directly: the trainer that produces this paper’s clean Top-K baseline (§3’s null model with its penalty set to zero) hardcoded the projection unconditionally; we added a config flag (default

Table 7: Six one-line implementation details and their individually-measured effect sizes, all significant at $5.03\text{--}5.16\sigma$ (13 seeds/arm). For reference, published architectural claims in this literature are typically reported without seed variance at all (§5.2); a $d > 10$ implementation effect is larger than most reported architectural ones.

Detail	Metric moved	$ d $
KL warmup schedule + bias parameterisation (combined)	liveness (loose)	13.0
Initial decoder scale	FVE	16.5
Decoder-gradient-projection omission	FVE	3.8–14.3*
Initial decoder-weight distribution (Gaussian vs. uniform)	FVE, frac. recovered	4.7, 7.3
F.relu(mu) presence/absence	liveness	15.3–19.3

*Reported as a single factor in two different comparison contexts (against different upstream generations of the ladder); both values are real, significant effects, not a range of uncertainty.

True, so every existing checkpoint’s behaviour is unchanged) and trained a new arm identical to the baseline in every other respect with the projection turned *off*, 13 seeds.

Table 8: Turning the same projection off in a plain, penalty-free Top-K SAE: null on every metric, both liveness thresholds agreeing (13 seeds/arm, 5.16σ floor).

Metric	projection on (baseline)	projection off	d (p)
FVE	0.900159 ± 0.0006	0.900009 ± 0.0005	$+0.3$ ($p = 0.50$)
frac. recovered	0.964336 ± 0.0005	0.964065 ± 0.0006	$+0.5$ ($p = 0.22$)
$< 0.1 \times k/d$	0.754169 ± 0.0013	0.753794 ± 0.0014	$+0.3$ ($p = 0.53$)
$< 0.5 \times k/d$	0.754357 ± 0.0014	0.754056 ± 0.0013	$+0.2$ ($p = 0.63$)

Nothing moves detectably, at the identical statistical power (13 seeds, 5.16σ floor) that detected $d = +3.8$ to $+16.5$ effects of the *same* one-line change inside the vSAE family. This answers the question directly, and narrows rather than weakens the claim Table 7 makes: the decoder-gradient-projection effect is real, large, and undocumented in either paper’s equations, but its scope is models carrying an activation penalty, not Top-K SAE training in general. A plain Top-K SAE and an independently-implemented plain Top-K SAE that differ only on this one line would not, on this evidence, be expected to differ measurably; a vSAE (or an L_2 -penalised Top-K SAE) and one that differs only on this line would. This is itself informative about *why* the effect exists: it is consistent with an interaction between the projection and the penalty term’s own gradient contribution, not with the projection mechanically mattering to decoder-norm renormalisation on its own, which would have shown up here too. (We did not find a below-threshold trend in either direction to further motivate that mechanism; the result is a clean null, not a smaller-but-present effect.)

5.2 A design-level point this implies

None of these effects required an unusual amount of statistical power to detect; they were detected at the same power (13 seeds/arm, $5.03\text{--}5.16\sigma$) used throughout this paper. The combinatorial floor on that power is worth stating plainly, because it caps what *any* design with fewer seeds can claim, regardless of the underlying effect size. A two-sided seed-permutation test’s minimum attainable p -value is $2/\binom{2n}{n}$ for n seeds per group (Table 9): at $n = 6$ that floor is $p = 0.00216$, i.e. 3.07σ — not because the effect is small, but because the design cannot express more evidence

than that, however large d actually is. Thirteen seeds per group is the first n at which this floor drops below the conventional 5σ threshold; at $n = 1$, the floor is $p = 1$ (0σ) — a single-seed design cannot express *any* level of statistical significance for a claim of the form “architecture A beats architecture B,” independent of how large the true effect is.

Table 9: The sigma ceiling of a two-sided seed-permutation test, by seeds per group. Same formula ($2/\binom{2n}{n}$) as every confirmatory comparison in this paper.

n/group	1	2	3	5	6	10	13	20
sigma ceiling	0.00	0.97	1.64	2.65	3.07	4.40	5.21	6.75

We checked how large this gap actually is by surveying every SAE-methods paper cited in this paper’s own related work (§8) — a sampling frame fixed by that bibliography, not curated for this subsection — for how many independently-seeded training runs each reports for a single hyperparameter configuration (Table 10). **Ten of ten report exactly one.** The one exception found anywhere in the survey — Bricken et al. [2023]’s second, independently-seeded transformer — is used only for a qualitative feature-universality check, not the paper’s main quantitative claims, and its own $n = 2$ sits below the conventional $p < 0.05$ threshold regardless. Several of these papers train *many* SAEs (SAEBench trains “over 200” [Karvonen et al., 2025]; JumpReLU trains “multiple 131k-width SAEs... of each type” [Rajamanoharan et al., 2024b]), but as a sweep over hyperparameters — width, sparsity, architecture — not as repeated seeds of one configuration; that breadth buys coverage across the hyperparameter space, not statistical power for any single comparison within it.

Table 10: Independently-seeded training runs per configuration, surveyed across every SAE-methods paper this paper cites. Full text of each paper searched via its arXiv HTML (two Anthropic/Transformer-Circuits posts exceeded a direct-fetch size limit; those two entries rest on multiple independent web searches quoting the papers’ own methodology instead, and are flagged as such).

Paper	seeds/configuration
Cunningham et al. [2023]	1
Bricken et al. [2023]	1 (2 for one supplementary check only)*
Templeton et al. [2024]	1*
Gao et al. [2024]	1
Rajamanoharan et al. [2024a]	1
Rajamanoharan et al. [2024b]	1
Bussmann et al. [2024]	1
Karvonen et al. [2025]	1
Marks et al. [2025]	1
Lu et al. [2025]	1

*Not backed by a direct full-text fetch; see caption.

One nuance is worth stating precisely rather than flattening: JumpReLU *does* report binomial confidence intervals for a manual human- interpretability rating study [Rajamanoharan et al., 2024b] — evaluation-level uncertainty, not training-seed-level uncertainty. This is exactly the distinction §A draws (a token- or rater-level test estimates evaluation noise and cannot support an

architecture-level claim) showing up unprompted in the surveyed literature: eval-level uncertainty quantification is not rare here; architecture-level (seed) uncertainty, this survey suggests, is close to absent.

This gap is not merely theoretical. Two papers outside the survey above — they measure seed *sensitivity* itself, not the field’s reporting practice, so they are not part of Table 10 — found substantial seed-to-seed instability in exactly the object a single-seed design cannot detect. Paulo and Belrose [2025] found only $\sim 30\%$ feature overlap between two otherwise-identical 131k-latent SAEs differing only in random seed (Llama-3-8B, replicated across multiple layers, LLMs, and datasets); Gerasimov et al. [2026] found individual features vary substantially across seeds in a large-scale multi-seed study, though they cluster into reproducible lower-rank subspaces. Neither result was designed to be detectable by any of the ten papers surveyed above, and both found it anyway once someone looked.

Every effect size in this paper is real in the sense that it survived a design capable of expressing 5σ ; the implementation choices catalogued in Table 7 are large enough that a design matching the field’s own modal practice ($n = 1$) could not, even in principle, distinguish them from the architectural effect a comparison was nominally about — not because those studies’ point estimates are wrong, but because a claim of the form “architecture A beats architecture B” is not a claim a single-seed design can license, independent of the true effect size.

6 A case study in metric disagreement: SCR and TPP on the same checkpoints

The preceding sections establish what the KL term and the reparameterisation trick do to reconstruction and to the dictionary’s own internal liveness statistics. A separate, harder question is whether any of this changes what downstream interpretability benchmarks report. We do not resolve this question in this paper; we report it as a worked example of a problem that generalises beyond this specific comparison.

6.1 The comparison and the first disagreement

Two recovered Pythia-70M checkpoints from the original preprint this codebase extends — a Top-K baseline ($d = 8192$, AuxK $\alpha = 0.03125$) and its corresponding vSAE ($\beta = 1$, fixed variance, AuxK *off*, 1,474/8192 live features) — were scored with SAEBench’s spurious correlation removal (SCR) and targeted probe perturbation (TPP) metrics [Karvonen et al., 2025] on the same dataset (`bias_in_bios`, professor/nurse) and the same size-response design: mask the baseline dictionary down to a grid of N features by usage frequency, score each point, and compare the vSAE’s own (unmasked, natural) score against the resulting curve at $N = 1,474$. SCR is SAEBench’s adaptation of the earlier SHIFT protocol [Marks et al., 2025] (ablate SAE latents a probe identifies as encoding a spurious attribute, and measure how much target-concept accuracy survives); TPP extends the same idea to simultaneous multi-class disentanglement.

Table 11: SCR and TPP disagree on the same two checkpoints, same dataset, same grid.

Metric	vSAE score	baseline curve at $N = 1474$ (<code>top_usage</code>)	margin
SCR	0.1017	0.0201	+0.082 (not explained by size)
TPP	0.1055	0.1905	−0.085 (explained by size)

SCR says the vSAE’s advantage over its baseline is not explained by dictionary size (its un-masked score clears every point on the masked curve, including the top-usage reference and a bracket of 90 random-subset draws). TPP says the opposite: the vSAE’s score sits *below* both the top-usage and the random-subset references at the same N . This is not a subtle effect that disappears under scrutiny; it is a sign flip on the exact same comparison, and it recurs in every diagnostic we ran against it.

6.2 Five diagnostics, none of which explains the disagreement away

We treated the disagreement as a claim requiring its own falsification attempt and ran five diagnostics, roughly cheapest-to-test first.

1. **Is the mean-across-thresholds summary hiding a shape change?** Both scorers’ internal per-threshold breakdown ($N \in \{2, 5, 10, 20\}$ ablated features) was extended and re-derived. SCR says “not explained” at all four thresholds; TPP says “explained” at all four, though decisively only at $N \leq 10$ (a near-tie at $N = 20$). The disagreement is not an artifact of averaging.
2. **Is it a test-set-noise artifact?** A 200-draw bootstrap of the cached test set (conditional on node effects fixed from the full train set) leaves both mean margins clearing zero in opposite directions: SCR $+0.082$ $[+0.024, +0.150]$, TPP -0.086 $[-0.090, -0.081]$. SCR’s interval is $\sim 15\times$ wider than TPP’s and, per-threshold, clears zero only at $N = 2$; a full bootstrap that also re-derives node effects from a train-set resample widens SCR’s interval further ($+0.088$ $[+0.008, +0.171]$) but still does not cross zero. The disagreement survives resampling, though the two sides are not equally sturdy.
3. **Do the two metrics select systematically different features?** A falsifiable prediction — SCR’s top-effect features should rank low in usage frequency, TPP’s should rank high, since disentangling a correlated pair plausibly needs a rare feature while separating many classes needs broad ones — was stated in advance and tested by reading which features each metric’s own effect computation ranks highest, cross-referenced against their usage-percentile rank. It is falsified: both metrics’ top features sit in the same 0.55–0.73 usage-percentile band at every N , the small gap between them flips sign between $N \leq 5$ and $N \geq 10$, and within-metric variance across classes dwarfs any between-metric difference.
4. **Does it generalise across datasets and class pairs?** Extending to three more `bias_in_bios` pairs and a second SAEbench dataset (Amazon reviews) finds both verdicts replicating as the dominant pattern, not reversing: SCR “not explained” on 7 of 8 (dataset, pair) points; TPP “explained” via its top-usage reference on both datasets, though its random-subset reference goes from decisive to a coin flip on the second dataset.
5. **Is the masked reference curve itself the wrong comparison?** Every point on the masked curve is the *same* baseline dictionary with entries zeroed post hoc; the vSAE’s 1,474 live features were learned together, and a dictionary trained from scratch at that exact size might organise its capacity differently. Training a real Top-K SAE at `dict_size = 1474` (otherwise identical config) and scoring it directly against the original professor/nurse point finds its SCR score (0.020127) matches the masked curve’s reference (0.020127) to 6×10^{-8} , and its TPP score (0.2025) is slightly *higher* than the masked reference (0.1905) — on that one point, this specific confound is not live. It is not this well-behaved everywhere: scoring the same trained-from-scratch checkpoint against the other 7 (dataset, pair) points item 4

added finds the reference-curve shift is often substantial for SCR (mean absolute shift 0.075, up to 0.138 at one point, comparable to the margins the verdict rests on) and flips one of eight points’ individual sign (Software/Electronics, $+0.110 \rightarrow -0.027$). **The aggregate verdict is essentially unmoved regardless:** the mean margin under the trained-small reference (+0.079) is marginally *larger* than under the masked reference (+0.072) on the same eight points, and 6 of 8 (was 7 of 8) still say “not explained.” TPP’s two available points (bias_in_bios, amazon; TPP has no per-pair concept) both keep their “explained by size” verdict under the trained-small reference. The honest summary is narrower than the single-point reading: the masking-vs-training confound is real and can move an individual pair’s verdict, but it does not systematically favour either metric’s side, so it does not explain the SCR/TPP disagreement itself. (A secondary finding en route, from the original point: even given the vSAE’s exact live-feature count as its budget, a plain Top-K network trained at that size keeps only 65.5% of it alive — matching a feature *count* does not guarantee matching how efficiently a network uses it.)

6.3 What we take this to show

We do not have an explanation for why SCR and TPP disagree here, and we are explicit that this section does not supply one. What we can report is that every mechanism we could think of to explain the disagreement away — an averaging artifact, sampling noise in the test set, a systematic difference in which features the two metrics reward, insufficient generalisation across data, and a masking-vs-training confound in the reference curve — fails to do so. One structural difference between the two metrics’ construction is at least consistent with SCR being the noisier of the two: SCR’s per-threshold score is a ratio (divided by `clean_acc` – `original_acc`, which can be small), while TPP’s is a plain difference, and SCR’s baseline curve is visibly flatter and noisier than TPP’s own equivalent curve across the same N range. This is consistent with, but does not establish, part of the disagreement being a metric-construction property rather than a property of the two dictionaries’ features — and it does not explain the sign flip itself, since SCR’s mean-level margin still excludes zero under both bootstraps.

We report this as a demonstration of a general problem rather than a finding specific to this vSAE. The same two checkpoints, the same dataset, the same day of measurement, and two metrics that are both standard, published, and widely used, disagree in direction on the same substantive question. Reporting only one of them — as is standard practice when a paper picks the metric that supports its narrative — would be a true statement about that one metric and a misleading one about the underlying claim. We regard the correct response to a finding like this as reporting the disagreement itself, with its own confound-checking, rather than resolving it by fiat or by omission.

Independent, more direct evidence has since appeared. Chanin [2026] audits SCR and TPP against three tests neither this paper nor SAEBench’s own release applied — reseed noise on fixed canonical SAEs, correlation with a synthetic benchmark’s known ground-truth quality, and discriminability across a training trajectory — and finds both fail basic sanity checks at their canonical settings: SCR and TPP *decline* over the course of training on the panels tested (implying an untrained SAE is “better” than a trained one), TPP’s canonical setting shows essentially zero correlation with ground-truth quality on synthetic SAEs, and SCR turns *negatively* correlated with ground-truth quality at large N , ranking a perfect oracle below 11 of 35 trained SAEs tested. The recommendation is to avoid both at their canonical settings entirely. This is a different, and more direct, kind of evidence than anything in this section — it audits each metric against a ground truth we do not have access to, rather than checking whether the two metrics agree with each

other — but it points the same way: neither metric’s number should be read as a settled fact about feature quality. It does not resolve which verdict this paper’s own SCR/TPP disagreement should be believed, if either; if anything, it weakens confidence in both, which is consistent with, not contrary to, the position this section already takes.

7 Discussion

What the three parts add up to. At fixed variance, adding a KL term to a Top-K SAE is adding an L_2 penalty and nothing else (§3); the empirical work needed to state that plainly is checking that two independent implementations of the identical objective actually agree, which they do only once five unrelated implementation choices are matched. Once sampling is switched on, the damage is large and mechanistically specific: it tracks how much the noise perturbs a hard selection mechanism’s chosen set, not some generic notion of representation quality, and that mechanism generalises across two architectures that scope their hard- k constraint differently but breaks down — in an informative way, not merely a noisier way — for a soft, gradient-learned threshold (§4). Both of these facts about the architecture are, in this codebase, larger than five one-line implementation details that appear in no equation the released code’s own paper states (§5). The composite claim is not “variational sparse autoencoders do not work” — it is that the standard construction (isotropic Gaussian prior, fixed unit variance, hard Top-K selection) cannot test the hypothesis it is usually offered in support of, because the noise it introduces interacts with hard selection in a way that is guaranteed to hurt and never guaranteed to help.

Constructive payload. If the goal is genuinely a stochastic sparse code, §4’s synthesis gives a specific, testable design recommendation rather than a negative result: put the noise in the *selection decision* (as Gumbel-Softmax and Concrete relaxations do for discrete choices generally [Jang et al., 2016, Maddison et al., 2016], and as Berthet et al. [2020]’s differentiable perturbed optimisers do specifically for top- k -type discrete selection — exactly the operation whose interaction with reparameterisation noise this paper measures — by perturbing the ranking *before* the argmax/top- k step and differentiating through the resulting expectation, rather than perturbing an already-selected value afterward), not in the magnitude of an already-selected feature. The hard-concrete stochastic gate of Louizos et al. [2018], developed for L_0 -regularised weight sparsity rather than sparse coding, is a close structural cousin: a continuously-relaxed *gate* decides membership, and a separate mechanism sets magnitude, which is exactly the separation this paper’s noise-in-selection recommendation asks for and JumpReLU’s own architecture (gate, then value) gestures toward without actually placing the *noise* at the gate. A prior with mass at zero — spike-and-slab being the textbook example for sparse codes, in contrast to the isotropic Gaussian’s zero mass at the origin, and already used for exactly this purpose in a VAE by Tonolini et al. [2020] — is a second, related direction: it would make the KL term itself do something other than shrink activation magnitude uniformly. Neither is tested here; both follow directly from identifying the specific mechanism this paper measures rather than only its symptom.

A note on the liveness–reconstruction relationship. An exploratory (not pre-registered) reading of every replicated arm in this battery found that reconstruction quality and feature liveness are not a single trade-off curve: pooled across 11 arms the two metrics look unrelated ($\rho = -0.14$, $p = 0.69$), but that null is two opposite-signed relationships cancelling — among the subset of arms that reach reasonable reconstruction quality, better reconstruction buys *more* near-dead features ($\rho = +0.86$, $p = 0.007$), the same direction the healthy end of the vSAE sweeps in this paper’s own

dose-response curves shows. We flag this only as context for interpreting Table 3–5’s FVE columns alongside their own liveness statistics, not as an additional pre-registered claim.

7.1 Limitations

- **Single model family for the controlled experiments.** All of §3–§5 runs on GELU-1L-scale residual stream activations at one layer, one dictionary size ($4 \times d_{\text{model}}$), one sparsity level ($k = 256$). Whether the decomposition’s null result and the dose-response’s coupling strength hold at larger model or dictionary scale is untested here.
- **The E4 headline comparison (§6) carries an uncontrolled confound of its own.** The recovered Pythia baseline trains with AuxK dead-feature revival on ($\alpha = 1/32$); the recovered vSAE trains with it off. AuxK is the standard dead-feature mitigation in this literature, so the headline “vastly fewer live features” comparison between these two specific checkpoints confounds the KL term with the absence of the standard remedy. This does not affect §4’s dose-response sweeps, which hold AuxK fixed throughout, but it does affect how strongly §6’s specific numbers should be read as characterising “the vSAE” versus characterising these two particular checkpoints.
- **The noise-scale sweeps (§4) are single-seed correlations at the arm level, not a permutation test.** $r = 0.9993, 0.9979, 0.9334/0.6297$ are Pearson correlations over six (or four) arm-level means, each itself averaged over 13 seeds; they are strong, visually unambiguous relationships, but the unit of statistical inference used elsewhere in this paper (a permutation test over seeds) does not apply to a six-point curve, and we do not claim σ -level significance for these correlations.
- **The JumpReLU result (§4) rests on eight grid points after confound control, four of them at reduced seed count.** The follow-up in §4.4.2 rules out $n = 4$ being simply too few points to trust, but the four new points ran at 5 seeds/arm rather than the main battery’s 13, and $n = 8$ arm-level points is still a small- n correlation, not a claim carrying the seed-permutation-test guarantees §A describes for the rest of this paper. An architectural (rather than soft-loss) L_0 cap would answer the original question more cleanly still, at the cost of turning JumpReLU into another top- k variant and defeating the point of testing it; not attempted here.
- **E4’s five diagnostics are single-seed, descriptive readings, not a confirmatory battery.** Unlike §3–§5, none of §6’s numbers come from a pre-registered multi-seed design; they establish that a disagreement exists and survives several specific attempts to explain it away, not that it would replicate at the same statistical standard as the rest of this paper.

8 Related work

Sparse autoencoders for language-model interpretability were introduced by Sharkey et al. [2022] and Cunningham et al. [2023], building on the superposition hypothesis of Elhage et al. [2022] and scaled by Bricken et al. [2023] and, to production-model scale, Templeton et al. [2024]. Top-K sparsity, which this paper’s baseline uses throughout, is due to Gao et al. [2024], who also introduce the AuxK dead-feature revival mechanism this paper’s §6 discusses as a confound. Gated SAEs [Rajamanoharan et al., 2024a], BatchTopK [Bussmann et al., 2024] and JumpReLU [Rajamanoharan et al., 2024b] are, respectively, the sparsity-penalty-side and (for the latter two) the

selection-mechanism-side alternatives this paper’s §4 and §5.2 draw on; Lu et al. [2025] is a recent critical re-examination of SAE evaluation practice in the same spirit this paper’s §6 and §5.2 are written in. SAEBench [Karvonen et al., 2025] is the source of the SCR and TPP metrics scored in §6, and of the sparse feature circuit evaluation methodology this codebase’s downstream tooling is built on [Marks et al., 2025]. §5.2’s survey of these ten papers’ own seed-count practice is corroborated by two independent studies of SAE seed sensitivity itself, neither part of the survey: Paulo and Belrose [2025] found only $\sim 30\%$ feature overlap between otherwise-identical SAEs differing only in random seed, and Gerasimov et al. [2026] found individual features vary substantially across seeds while clustering into reproducible lower-rank subspaces. The variational autoencoder and its reparameterisation trick [Kingma and Welling, 2013] and the β -weighted KL trade-off this paper’s Eq. 2 follows [Higgins et al., 2017] are the two pieces of machinery this paper’s central object combines with Top-K selection. The straight-through gradient estimator used to make JumpReLU’s threshold trainable [Bengio et al., 2013], the two continuous relaxations of discrete sampling this paper’s constructive discussion points toward as the right place to put stochasticity in a selection-based architecture [Jang et al., 2016, Maddison et al., 2016], and, closer still to top- k selection specifically, the differentiable perturbed optimisers of Berthet et al. [2020] and the hard-concrete stochastic sparsity gate of Louizos et al. [2018] — both of which place noise at the discrete decision itself rather than at an already-selected value’s magnitude, and the latter of which structurally separates a gate from a magnitude the way JumpReLU’s own architecture does, without placing noise at the gate — are the closest prior treatments of the specific tension §4 measures: that a continuous relaxation of a discrete choice and a discrete choice made from continuously-perturbed inputs are not the same operation, and only one of them is what a sampled sparse code actually needs. Tonolini et al. [2020]’s spike-and-slab VAE is the closest prior instance of this paper’s second constructive suggestion (a prior with mass at zero), applied to a different sparse-coding setting than Top-K selection. We are not aware of a prior controlled decomposition of a variational sparse autoencoder into its constituent implementation choices, or of a dose-response measurement of the reparameterisation-selection interaction across multiple selection mechanisms. Chanin [2026] independently audits SCR and TPP against ground-truth and training- trajectory sanity checks and recommends against both at their canonical settings; §6 is, to our knowledge, the first report of the two metrics giving opposite-signed verdicts on the same checkpoints, which is a different, complementary kind of finding to Chanin’s — disagreement between two metrics versus failure of either against a ground truth.

9 Conclusion

Adding a KL term to a Top-K sparse autoencoder, at the fixed variance every public implementation we examined actually trains with, adds an L_2 activation penalty and nothing else — verified not merely as algebra but as an empirical equivalence between two independently-optimised implementations, once the five unrelated details that initially separated them are identified and matched. Once sampling is switched on, the resulting damage has a specific, measurable mechanism — selection instability under a hard cardinality constraint — that generalises across two architectures scoping that constraint differently and fails informatively, rather than simply more noisily, for a soft alternative. Both findings required an amount of implementation-level control that a typical single-seed, single-implementation SAE comparison does not have, and the implementation details that turned out to matter are, individually, larger effects than many of the architectural claims this literature reports without seed variance at all. We close with a case where our own confound-checking did not resolve a disagreement between two standard metrics, offered in the spirit the rest

of this paper is written in: a claim is only as good as the specific alternative explanations that were checked and ruled out, and absent that checking, a result this literature currently accepts as supporting evidence for “variational structure helps” is, as far as we can tell, evidence for none of the things it is usually cited for.

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A Statistical methodology

Every seeded comparison in §3–§5 is a two-sided permutation test over training seeds (not tokens, not checkpoints): the null hypothesis is that arm labels are exchangeable across the $2n$ seeds in a comparison, and the test statistic is the observed difference in means. Two combinatorial ceilings bound what such a test can report, independent of effect size, and every result in this paper states which one applies.

The seed-count floor. For n seeds per group, the minimum attainable two-sided p -value under exact enumeration is $2/\binom{2n}{n}$. At $n = 6$ this floor is $p = 0.00216$ (3.07σ); no effect, however large, can be reported below this floor at $n = 6$. At $n = 13$ the floor drops to $p = 2.5 \times 10^{-7}$ (5.16σ) once Monte Carlo sampling (below) is accounted for, which is why every confirmatory result in this paper is reported at 5.03 – 5.16σ rather than an arbitrarily large significance level: this is the design’s own ceiling, not a property of any particular effect.

The Monte Carlo floor. Above $\binom{2n}{n} = 200,000$ possible assignments ($n > 9$ or so, depending on group balance), exact enumeration is replaced by Monte Carlo sampling of the permutation distribution, whose own floor is $1/(n_{\text{perm}}+1)$ under the conservative estimator $(\text{count}+1)/(n_{\text{perm}}+1)$ (the naive $\text{count}/n_{\text{perm}}$ estimator is anti-conservative and can report $p = 0$). At the common default of $n_{\text{perm}} = 100,000$ this caps evidence at 4.42σ *regardless of effect size*; every result in this paper instead uses $n_{\text{perm}} = 4,000,000$ (a vectorised implementation makes this take under one second), giving a floor of 5.16σ .

The two-threshold liveness rule. Feature liveness is reported at two pre-registered thresholds, $< 0.1 \times k/d$ and $< 0.5 \times k/d$ of the architecture’s own expected per-feature firing rate, with the rule fixed in advance: a result counts as robust only if both thresholds agree in direction; disagreement between them is treated as a positive finding (a shape change in the usage distribution, not a location shift) rather than resolved by picking whichever threshold supports the narrative. This rule caught a genuine shape change — not a location shift — twice in the course of this project’s larger confirmatory battery (§3’s init factor, and one comparison not otherwise reported in this paper), in both cases exactly the failure mode a single threshold would have silently reported as a clean one-directional effect.

B The full E1 code diff

Every difference between `top_k.py` and `vsae_topk.py` (plus the two training scripts and a third trainer, `top_k_with_feature_penalty.py`, that differs from `top_k.py` only by the penalty term), enumerated by direct code reading and frozen before §3’s closing arm ran.

C Bugs found and fixed in the course of this work

Three checkpoint-affecting bugs and one training-blocking bug set were found and fixed while producing the results in this paper; we report them because each one silently altered a reported number before it was found, and because the corrections change how earlier readings of this same data should be interpreted.

Table 12: The frozen 15-item code diff behind §3.

Status	Item(s)
Matched by config (2)	<code>kl_warmup_steps</code> ; bias parameterisation
Run as a measured factor (2)	decoder init scale; gradient-projection omission
No-op, measured (2)	dead-feature-liveness accounting convention; the ± 10 pre-selection clamp (never binds: max observed activation 0.194 against a clamp at 10)
No-op, static (7)	<code>abs()</code> in selection; an unused <code>threshold</code> buffer; the geometric-median initialisation guard; two epsilon/clamp guards; the dead-feature counter update site; a 2.7% initial-norm offset
Not a factor by design (1)	RNG consumption order — the seed itself is the unit of randomisation the permutation design already treats as noise
Run as the closing factor (1)	initial decoder-weight distribution (normalised Gaussian vs. normalised uniform) — predicted null, found not null ($d = -4.7$ on FVE)

A checkpoint-save bug corrupted every saved learned-variance bias. The routine that rescales a trained model’s biases from normalised-activation space back to raw-activation space at save time (correct for the encoder and decoder biases) was, until fixed, *also* applied to the log-variance encoder’s bias — mathematically wrong, since a log-variance is not on the additive scale that rescaling assumes, and it drove every saved learned-variance checkpoint’s sigma to read as fully clamp-collapsed regardless of what was actually learned. Corrected (the log-variance encoder’s *weight* is rescaled instead, which is scale-homogeneous for this quantity), the learned sigma at convergence is a moderate $\sigma \approx 0.27$, not the clamp floor $\sigma = 0.0498$ — the posterior does not collapse, contrary to an earlier, uncorrected reading of the same checkpoints. This correction does not change §4’s dose-response results, which fix σ by its *initial* value rather than reading a learned one, but it does change the interpretation of §3–§4’s $\beta = 0$ control’s own learned sigma, and an identical bug was independently found and fixed in the BatchTopK trainer while preparing §4’s BatchTopK sweep.

The official re-evaluation pipeline was checked against an uncommitted proxy, and the proxy did not replicate. An early estimate of how much the bias-correction bug (above) affected officially-reported reconstruction quality, computed with a standalone proxy script, suggested a large (≈ 0.12 FVE) correction. Re-running the actual evaluation pipeline on every affected checkpoint with the bug corrected found the real effect is an order of magnitude smaller (≈ 0.005 FVE); the proxy’s number does not survive contact with the pipeline it was estimating, although the qualitative claim it was built to support (the posterior does not collapse) does. Every number reported in §3–§4 of this paper is from the official pipeline, not the proxy.

Three bugs in the JumpReLU trainer, one of them structural. As described in §4: a dead gradient on the threshold parameter (a non-differentiable comparison where the codebase’s own non-variational JumpReLU trainer already used a straight-through estimator for exactly this reason); a missing L_0 -target sparsity loss term (added, mirroring the non-variational trainer’s own);

and a gate applied before sampling rather than after, which would have made this section’s central measurement (selection churn under noise) structurally impossible to observe at all, since a gate computed before noise is added can never change which features are selected, only perturb the magnitude of ones already chosen. The third is a genuine design fork rather than an unambiguous bug — gating before sampling is a valid architecture, just a different one, one that cannot test what §4 tests — and was resolved by restructuring to match the encode-then-select order the other two architectures in §4 use, so that all three architectures in that section answer the same question. A fourth, independent bug (decoder renormalisation not rescaling the learned threshold alongside it, silently corrupting evaluated output on any non-unit-norm decoder) was also found and fixed.

D Reproducibility

All arms in this paper are defined declaratively in this codebase’s `falsification/run_arm.py` and trained with `python falsification/run_arm.py --arm <name> --seed <seed>`; `falsification/preflight.py` validates every arm’s wiring without requiring a GPU. By section:

- §3: `e1_penalty`, `e1_vsae_ref`, `e1_vsae_ref_unitinit`, `e1_vsae_ref_gradproj`, `e1_vsae_ref_fullmatch`.
- §4: `a2_sigma_init_{m1,m3,m4,m5}` plus `e2_sampling_only` and `e2_sigma_low_init` (TopK); `a3_batchtopk_*` (BatchTopK); `a4_jumprelu_*` (JumpReLU).
- §5: the ReLU factor is `e3_masked_kl / e3_masked_kl_relu`; the projection-generalisation control (§5.1) is `baseline` vs. `claim4_baseline_noproj`.

Every arm ran at 13 seeds, except §4.4.2’s four densifying JumpReLU points (`a4_jumprelu_sigma_init_{m3_5,m4_5,m5_5,m6}`), which ran at 5 (`falsification/read_a4_followup.py`). The falsification test suite (`python -m pytest falsification/tests/ -q`) exercises the permutation-test implementation itself, including the p -value floor formulas in §A, and stayed green throughout this project.